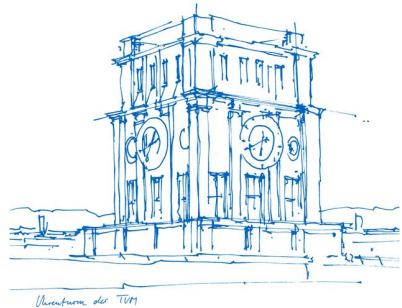


FineFlow–Espresso: A Calibratable 1D Model of Fines Migration

**A continuum model linking fines transport
and hydraulic resistance**

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Presentation outline

- Motivation and research question
- Physical mechanisms and model scope
- Governing equations and closures
- Numerical solution and verification
- Illustrative results and model sensitivity
- Calibration and validation strategy
- Conclusions and next steps

Section 1 of 7

Motivation and research question

How do fines change resistance and cup washout?

Motivation and research question

Can a 1D model reproduce the experimental response?

- Predict the outlet flow rate for the measured pressure drop.
- Predict the amount of fines reaching the cup.

How does fines redistribution change the hydraulic resistance and cup washout?

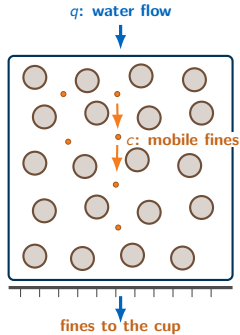
Darcy relates pressure to flow through permeability. Deposits change pore space and resistance. Release, capture and remobilization determine the fines distribution.

Section 2 of 7

Physical mechanisms and model scope

Release, capture and remobilization inside the puck

Flow and fines inside the coffee bed



Flow and mobile fines

- Q : flow rate; A : puck area; $q = Q/A$: superficial velocity;
- c : mobile-fines concentration; $q c$: advective fines flux.

Pore structure and retained fines

- ε : porosity; K : permeability;
- m_a : remaining initially attached fines;
 m_d : recaptured deposits;
- m_a, m_d : mass per bulk volume; c : mass per water volume.

Distinct retained populations assume different kinetics or hydraulic effects.

Scope of the 1D continuum model

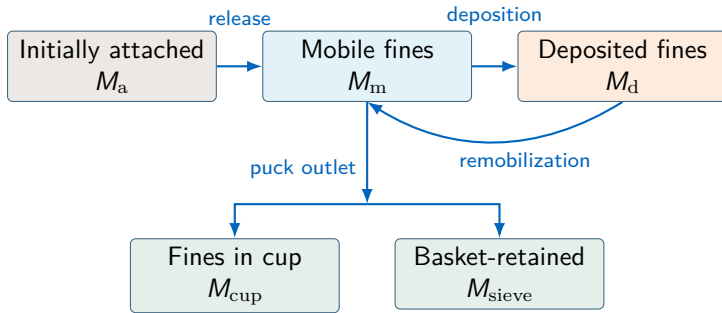
Present approach

- Forward flow with cross-section-averaged fields varying in z and t .
- CT stage snapshots can constrain axial structural changes.
- Pressure and flow histories constrain the evolving total resistance.

Limits and next measurements

- **Reverse flow and radial channeling are outside the scope of this work.**
- CT shows radial heterogeneity. Measurements must test whether axial averaging captures the overall hydraulic response.
- Time-resolved cup-fines data are a proposed measurement; availability is unconfirmed.
- Pore networks and 2D/3D continuum models are future extensions.

Fines exchange locally and leave through the outlet



$$M_0 = M_a + M_m + M_d + M_{sieve} + M_{cup}.$$

$$M_a = A \int m_a dz, M_m = A \int \varepsilon c dz, M_d = A \int m_d dz.$$

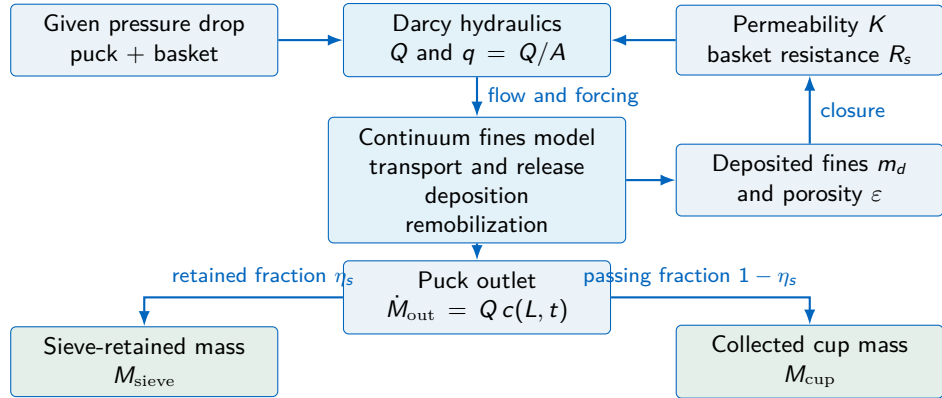
Accounting domain: puck + basket + collected cup fines.

Section 3 of 7

Governing equations and closures

Balances and constitutive choices for fines and flow

Model overview: pressure, flow and fines reaching the cup



Kozeny–Carman uses $\varepsilon(m_d)$ to obtain K . The exponential alternative uses m_d directly.

Darcy flow and the permeability closure have different roles

Darcy links the measured pressure drop to outlet flow

$$-\frac{\partial p}{\partial z} = \frac{\mu q}{K(z, t)}, \quad q = Q/A.$$

Lower permeability K means greater resistance and less flow at the same pressure drop.

$$R_{\text{tot}} = \frac{\mu}{A} \int_0^L \frac{dz}{K(z, t)} + R_s, \quad Q = \frac{\Delta p_{\text{total}}}{R_{\text{tot}}}.$$

R_s : basket hydraulic resistance [Pa s m^{-3}]. μ : water viscosity. Δp_{total} : pressure drop across puck plus basket.

Next: how do deposits change K ?

The permeability closure describes that structural response. Fines balances then determine how the deposits evolve.

Two alternative closures for the same permeability

Both closures supply K to Darcy. Deposits reduce the available pore space.

$$\varepsilon = \max(\varepsilon_0 - m_d/\rho_d, \varepsilon_{\min})$$

ρ_d : deposit density. Capture raises m_d and remobilization lowers it. Initial release affects porosity indirectly through later capture in this model.

Kozeny–Carman: structural baseline

$$\frac{K}{K_0} = \left(\frac{\varepsilon}{\varepsilon_0}\right)^3 \left(\frac{1 - \varepsilon_0}{1 - \varepsilon}\right)^2$$

Relates permeability to porosity under fixed effective particle scale and Kozeny factor.

This simplified relation may miss changes in pore connectivity and localized plugging.

Exponential: empirical comparison

$$\frac{K}{K_0} = \exp\left(-\beta \frac{m_d}{m_{\text{cap}}}\right)$$

Assumes a direct sensitivity to deposit loading. β controls the imposed blockage strength.

This empirical law has no coffee-data calibration here.

Identical initial K_0 . Neither closure has experimental validation.

Why the illustrative model used an exponential law

Motivation: represent strong blockage with a simple adjustable law

The hypothesis was that fines obstruct narrow flow paths, allowing a large resistance increase with a modest change in total porosity.

The exponential form stays positive and decreases smoothly. These useful mathematical properties do not establish physical validity.

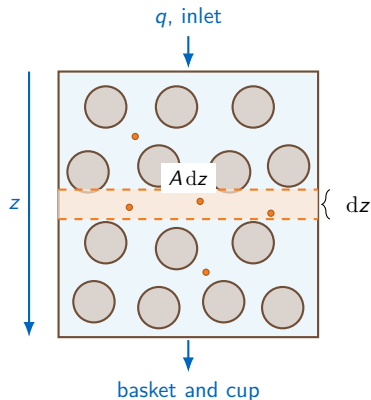
m_d/m_{cap}	K/K_0 at $\beta = 5.5$
0	1
0.5	0.064
1	0.0041

An illustrative assumption

Neither the exponential law nor $\beta = 5.5$ was fitted to the coffee data or taken from a cited coffee study. Values shown are before any numerical lower bound.

Strong blockage is substantially imposed by this closure; it is not independent evidence of the mechanism.

Continuum balance over an infinitesimal slice



Water-filled pore space

$$dM_m = \varepsilon c A dz$$

Mobile fines cross the slice boundaries by advection and dispersion.

Stationary coffee skeleton

$$dM_a = m_a A dz, \quad dM_d = m_d A dz$$

Release and capture exchange fines locally between water and skeleton.

The highlighted slice defines the continuum balance. It is not a numerical cell.

Continuum equations and optional surface loading

$$\frac{\partial(\varepsilon c)}{\partial t} + \frac{\partial(qc)}{\partial z} = \frac{\partial}{\partial z} \left(\varepsilon D \frac{\partial c}{\partial z} \right) + R_{\text{rel}} - R_{\text{dep}} + R_{\text{rem}},$$

$$\frac{\partial m_a}{\partial t} = -R_{\text{rel}}, \quad \frac{\partial m_d}{\partial t} = R_{\text{dep}} - R_{\text{rem}}.$$

Equivalent surface description

$$m_a = \sigma c_a, \quad m_d = \sigma c_d, \quad [\sigma] = \text{m}^2/\text{m}_{\text{bulk}}^3, \quad [c_a] = [c_d] = \text{kg}/\text{m}^2.$$

Use $\partial_t(\sigma c_a)$ if the specific surface area changes.

Only mobile fines travel axially. All R terms have units $\text{kg}/(\text{m}_{\text{bulk}}^3 \text{ s})$.

Release and capture are constitutive hypotheses

Release of initially attached fines

$$R_{\text{rel}} = k_{\text{rel}} m_a (|q|/q_{\text{ref}})^{a_r}, \quad [k_{\text{rel}}] = \text{s}^{-1}.$$

Capture of mobile fines

$$R_{\text{dep}} = k_{\text{dep}} \chi(z) |q| c \left[1 - \frac{m_d}{m_{\text{cap}}} \right]_+, \quad [k_{\text{dep}}] = \text{m}^{-1}.$$

- Release depends on the remaining available inventory and flow.
- Capture depends on mobile flux and available deposit capacity.
- $\chi = 1$ is the unbiased baseline. Axial weighting needs evidence.
- Compare these laws using multiple flow rates and spatial deposit data.

Deposit remobilization and local pore accessibility

Deposits return to the water only above a forcing threshold

$$R_{\text{det}}^* = k_{\text{det}} m_d \left[\frac{G}{G_{\text{crit}}} - 1 \right]_+^b, \quad G = \left| \frac{\partial p}{\partial z} \right|.$$

At $G \leq G_{\text{crit}}$, remobilization is zero. Above the threshold, more deposits and stronger forcing increase the potential detachment rate.

$$R_{\text{rem}} = P_{\text{mob}} R_{\text{det}}^*, \quad P_{\text{mob}} = \frac{1}{1 + \exp[s(d_f/d_h - \lambda_c)]}.$$

- P_{mob} : fraction that can enter flowing pores locally.
- Larger fines d_f relative to pore diameter d_h reduce this fraction.
- λ_c : size ratio at 50% access. s : transition steepness.
- The remaining fraction stays deposited. Mobile fines may deposit again.

Notation: $R_{\text{rem}} = R_{\text{wash}}$ and $P_{\text{mob}} = P_{\text{esc}}$ in the code. The logistic law is an unvalidated closure.

Fines leaving the puck: sieve retention and cup collection

The outlet carries mobile fines out of the puck

$$\dot{M}_{\text{out}}(t) = Q(t)c(L, t) \quad [\text{kg/s}].$$

$c(L, t)$ is the fines concentration at the puck outlet. The outlet model neglects dispersive exchange with the basket.

Retained by the sieve

$$\dot{M}_{\text{sieve}} = \eta_s \dot{M}_{\text{out}}$$

The basket retains the fraction η_s of the outgoing fines.

$$M_j(t) = \int_0^t \dot{M}_j(\tau) d\tau, \quad j \in \{\text{out}, \text{sieve}, \text{cup}\}, \quad M_{\text{out}} = M_{\text{sieve}} + M_{\text{cup}}.$$

Zero initial collected masses. The illustrative runs use constant $\eta_s = 0.35$, so 65% reaches the cup.

Retention η_s requires calibration. It differs from hydraulic resistance R_s . In the current model, sieve loading does not change R_s .

Collected in the cup

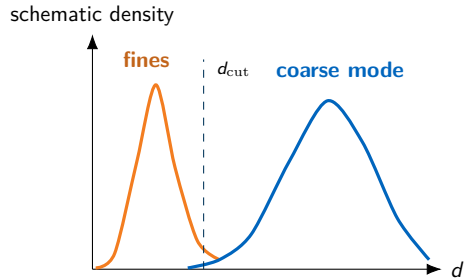
$$\dot{M}_{\text{cup}} = (1 - \eta_s) \dot{M}_{\text{out}}$$

The remaining fraction passes through the basket into the cup.

The PSD is reduced to quantities used by the 1D closures

Measured bimodal PSD

- Fit a five-parameter mixture model.
- Choose a mobility cutoff d_{cut} .
- Estimate the fine fraction and its releasable share.
- Calculate representative fine and coarse diameters.



Use in the model: the fine mass fraction and its releasable share set the initial attached inventory $m_{a,0}$. The fine diameter d_f controls local pore access. A representative bed diameter helps estimate the initial pore size.

Source: Bora and Briesen, Journal of Food Process Engineering 49 (2026), e70645.

Constant pressure or PI control

Constant pressure: controlled reference

$$\Delta p_{\text{total}} = 9 \text{ bar}, \quad Q = \Delta p_{\text{total}} / R_{\text{tot}}(t).$$

The source supplies whatever flow maintains 9 bar. This represents an experimental 9 bar plateau only when the machine can sustain that pressure.

Flow-limited PI: machine–puck prediction

$$e = p_{\text{set}} - \Delta p_{\text{total}}, \quad Q_{\text{command}} \leq Q_{\text{max}}.$$

The controller adjusts pump flow, with feedforward and pump lag. A permeable puck may stay below the target because the pump reaches its flow limit.

PI represents machine constraints. Its predictive value requires representative pump limits and controller settings.

Pressure reference and experimental comparison

Our priority: validate flow and cup fines using measured pressure

- Prescribe the actual pressure drop to test the puck model without fitting uncertain machine-control parameters.
- Constant 9 bar is the reference for a sustained 9 bar interval. Use the actual pressure for experiments below the target.
- Retain flow-limited PI to predict pressure and flow for different initial PSDs, with a representative machine model.

$p_{\text{out}} = 0$, $p_{\text{puck exit}} = QR_s$. The measured pressure must refer to the same endpoints as the model.

Measured pressure for validation. PI for machine–puck prediction. Constant 9 bar for comparison.

Measured time-series input remains an extension. The current saturated-puck model excludes initial wetting.

Boundary conditions for clean feed and an open outlet

$$J = qc - \varepsilon D \partial_z c$$

Inlet: no fines supplied

$$J(0, t) = qc_{\text{in}} = 0$$

The feed is clean water. We prescribe the total incoming fines flux.

This does not require imposing both $c = 0$ and $\partial_z c = 0$ on the interior solution.

In the code, boundary dispersive fluxes are omitted; inlet total flux is zero and outlet flux is qc . Reconsider this closure if downstream mixing or back-dispersion matters.

Outlet: convective removal

$$\partial_z c(L, t) = 0, \quad J(L, t) = qc(L, t)$$

We neglect dispersive exchange with the downstream basket/cup region.

This open-outlet approximation allows fines to leave by advection, without a modeled downstream concentration field.

Section 4 of 7

Numerical solution and verification

Algebraic Darcy hydraulics and finite-volume fines transport

Darcy flow follows from the current hydraulic resistance

At each update, use the current cell permeabilities $K_i(t)$

$$R_{\text{puck}} = \frac{1}{A} \int_0^L \frac{\mu}{K(z, t)} dz \approx \frac{\mu}{A} \sum_i \frac{\Delta z_i}{K_i}, \quad R_{\text{tot}} = R_s + R_{\text{puck}}.$$

Operating mode	Algebraic hydraulic evaluation
Pressure imposed	$Q = \Delta p_{\text{total}} / R_{\text{tot}}$
Flow imposed	$\Delta p_{\text{total}} = QR_{\text{tot}}$
PI control	Pump dynamics supply $Q(t)$, then $\Delta p_{\text{total}} = Q(t)R_{\text{tot}}$.

R has units Pa s/m^3 . The cell sum evaluates the resistance integral.

Darcy hydraulics use algebraic relations. FVM advances the fines-transport balance.

Velocity and pressure profiles follow from Darcy

Velocity from the common flow rate

$$q(t) = \frac{Q(t)}{A}, \quad u_{\text{pore},i}(t) = \frac{q(t)}{\varepsilon_i(t)}.$$

q is the superficial velocity, uniform along the constant-area puck.

The local pore velocity changes with porosity.

Pressures are relative to the cup outlet. The recurrence gives face pressures, with cell pressure at the midpoint for a cellwise constant gradient.

Pressure from the local resistance

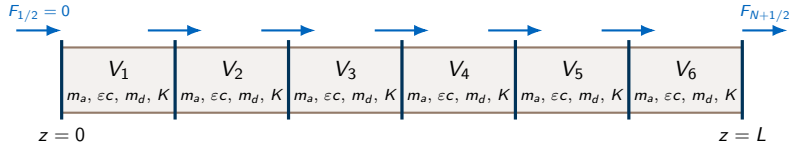
$$G_i = \frac{\mu q}{K_i}, \quad p(L) = QR_s.$$

$$p_{i-1/2} = p_{i+1/2} + G_i \Delta z_i.$$

Starting at the basket, accumulate each cell's pressure loss upstream.

Numerical quadrature reconstructs pressure. No separate momentum or pressure PDE is solved.

The finite-volume method enforces flux balance cell by cell



$$\frac{(\epsilon c)_i^{n+1} - (\epsilon c)_i^n}{\Delta t} = -\frac{F_{i+1/2} - F_{i-1/2}}{\Delta z} + R_{\text{rel},i} - R_{\text{dep},i} + R_{\text{rem},i}.$$

$$F_{i+1/2} = \underbrace{q c_i}_{\text{advection}} - \underbrace{\epsilon_{i+1/2} D \frac{c_{i+1} - c_i}{\Delta z}}_{\text{dispersion}}, \quad q \geq 0.$$

First-order upwind advection and centred dispersion. No high-order flux limiter.

Shared face fluxes cancel between adjacent cells, preserving the global mass balance.

Why finite volumes for fines transport?

The main requirement is consistent mass accounting

- Each cell stores a fines inventory and exchanges mass through its faces.
- One shared face flux leaves one cell and enters the next.
- Summing the cells cancels internal fluxes; boundary fluxes give collected fines.
- Local release, capture and remobilization transfer mass between inventories.

Alternative	Assessment for this model
Finite differences	Equally viable in conservative flux form; on a uniform grid, they can give the same update.

FVM is a transparent first implementation. Its first-order upwind diffusion requires grid refinement.

The smallest timescale limits the numerical step

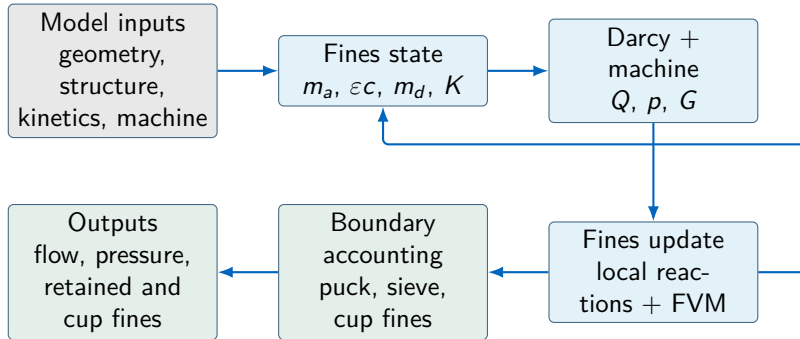
$$\Delta t = \min(\Delta t_{\max}, \Delta t_{\text{transport}}, \Delta t_{\text{pump}}, t_{\text{end}} - t)$$

$$\Delta t_{\text{transport}} = \frac{0.42}{|q|/(\varepsilon_{\min} \Delta z) + 2D/\Delta z^2}$$

Limit	Purpose
$\Delta t_{\max}$	User-selected upper bound for temporal resolution.
$\Delta t_{\text{transport}}$	Combined explicit advection–dispersion restriction. Here $\varepsilon_{\min} = \min_i \varepsilon_i(t)$.
$\Delta t_{\text{pump}} = \tau_p/4$	Used only in PI mode to resolve pump response; this is not the PI integral time.
$t_{\text{end}} - t$	Stops exactly at the requested final time.

For $D = 0$: $\text{Co} = |q|\Delta t/(\varepsilon_{\min} \Delta z) \leq 0.42$. Timestep refinement still checks accuracy of kinetics and coupling. The transport bound is the implemented estimate, not a general heterogeneous-field proof.

Coupled physics and its numerical update



Physical coupling: deposit changes ε , K and d_h ; these change pressure and flow; the new hydraulics then alter release, deposition and washout.

Quasi-steady flow is a physical assumption. Update ordering is a numerical choice.

Numerical verification and physical validation

Verification

- Conservation and nonnegative inventories.
- Analytical hydraulic limiting cases.
- Grid and timestep convergence.
- Controller limits and boundary accounting.

Validation

- Pressure and flow measurements.
- CT structural changes.
- Collected fines and size distributions.
- Predictions for withheld cases.

Current unit tests: 17/17 passed. The four pressure-mode runs conserve fines mass to relative error below 2.1×10^{-15} . Physical validation remains outstanding.

Section 5 of 7

Illustrative results and model sensitivity

How strongly do the hydraulic and permeability choices matter?

Darcy gives practically the same results

We ran the same case twice, changing only the flow law.

Result at 60 s	Darcy–Forchheimer	Darcy
Pressure [bar]	8.9906	8.9906
Flow rate [mL/s]	0.04535	0.04535
Cup fines [g]	0.03993	0.03993

Values rounded for display.

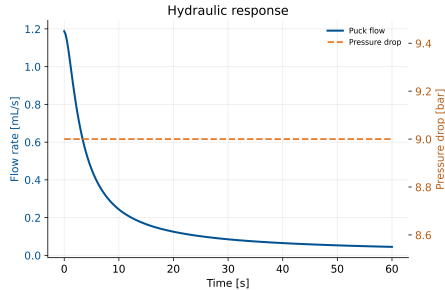
Across the entire simulation, the largest pressure difference was only **0.0203 Pa**.

We therefore use Darcy for this tested case.

Controlled 60 s simulations with identical exponential permeability, kinetics and PI settings.

The illustrative case develops strong viscous blockage

Constant 9 bar across puck + basket. Exponential permeability.



At 60 s

- total pressure: 9.00 bar;
- flow: 0.0447 mL/s;
- minimum K/K_0 : 0.0125.

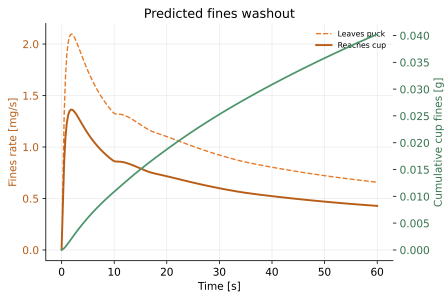
Interpretation

At fixed 9 bar, deposition increases resistance and reduces flow. No PI controller is needed.

New Darcy baseline: exponential permeability, constant 9 bar. Illustrative parameters, no experimental calibration.

The model distinguishes puck washout from fines in the cup

Constant 9 bar across puck + basket. Exponential permeability.



Cumulative masses

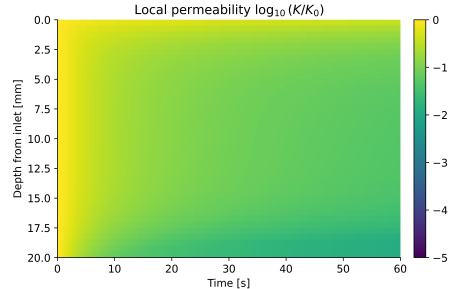
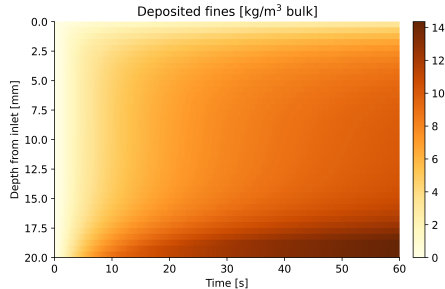
- puck outlet: 0.0619 g;
- basket retained: 0.0217 g;
- cup: 0.0402 g.

With $\eta_s = 0.35$, the basket retains 35% of the fines that leave the puck.

Constant 9 bar Darcy baseline with exponential permeability; constant basket retention.

The axial fields locate the effective blocking region

Constant 9 bar across puck + basket. Exponential permeability.

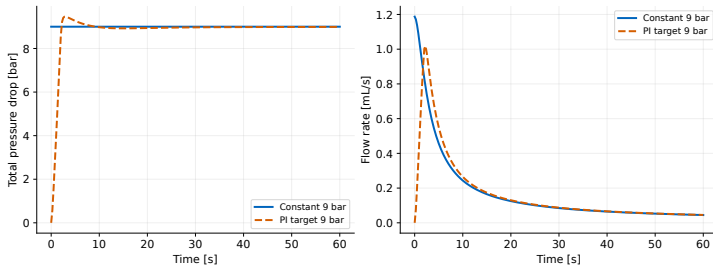


These fields can be compared with interrupted CT reconstructions, but a 1D layer cannot represent lateral channel geometry.

Constant 9 bar Darcy baseline with exponential permeability. The later closure comparison retains its original PI settings.

Fine puck: constant pressure versus PI

Constant 9 bar vs. PI, 9 bar target. Exponential permeability.

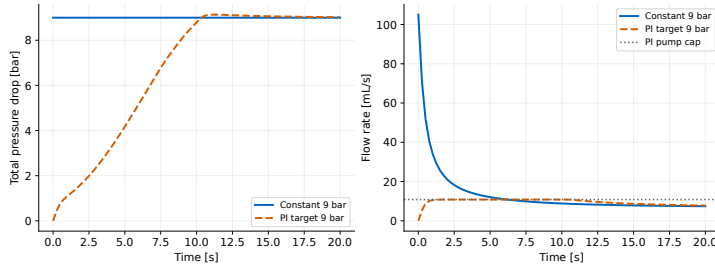


At 60 s: constant pressure gives $Q = 0.0447$ mL/s; PI gives $Q = 0.0454$ mL/s. The PI startup ramp changes the earlier fines history.

Matched 60 s runs. Only control mode changes. Darcy + exponential, illustrative parameters.

Coarse puck: the pump cap limits initial pressure

Constant 9 bar vs. PI, 9 bar target. Exponential permeability.

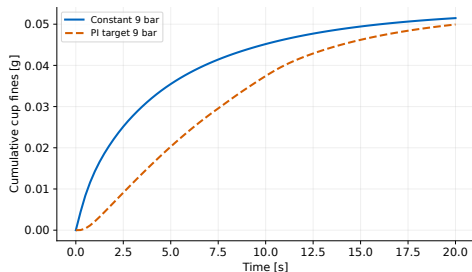


Constant 9 bar initially requires 105 mL/s, above the PI cap of 10.83 mL/s. PI reaches only 1.66 bar at 2 s. Deposition later raises resistance enough to reach the target.

Matched 20 s runs with active fines. This coarse case does not stay below 9 bar throughout. No experimental calibration.

Pressure history also changes fines reaching the cup

Constant 9 bar vs. PI, 9 bar target. Exponential permeability.



Same coarse puck and kinetics

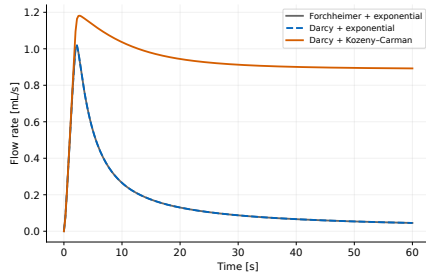
Different pressure histories change flow, release and deposition.

A fixed 9 bar run cannot directly represent a lower-pressure experiment.

Only control mode differs. Basket retention and permeability closure are identical.

Permeability closure strongly changes the flow response

Flow-limited PI, 9 bar target. Exponential vs. Kozeny–Carman.



At 60 s

Exponential:

$$Q = 0.0454 \text{ mL/s}$$

Kozeny–Carman:

$$Q = 0.892 \text{ mL/s}$$

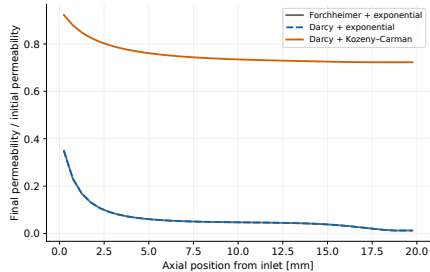
Both Darcy runs approach the 9 bar PI target.

The closure changes resistance and flow, even with nearly the same final pressure.

Same initial K_0 , kinetics and machine parameters. Gray and blue curves overlap: inertia is negligible.

Kozeny–Carman predicts weaker permeability loss

Flow-limited PI, 9 bar target. Exponential vs. Kozeny–Carman.



Minimum K/K_0

Exponential:

0.0126

Kozeny–Carman:

0.723

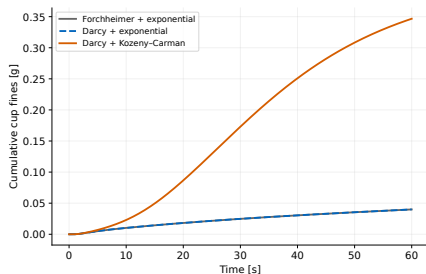
The same deposit-to-porosity law feeds both closures.

The exponential law imposes stronger blockage. The comparison does not establish which closure is correct.

Final profiles at 60 s. Coupled deposit fields also differ because flow feeds back on kinetics.

The permeability closure also changes cup fines

Flow-limited PI, 9 bar target. Exponential vs. Kozeny–Carman.



Cup fines at 60 s

Exponential:

0.0399 g

Kozeny–Carman:

0.3466 g

Same basket retention:

$\eta_s = 0.35$.

This is closure sensitivity, not validation. Neither prediction is an experimental target.

All kinetic and machine parameters held fixed. No fitting to reproduce the exponential-law result.

Section 6 of 7

Calibration and validation strategy

Which measurements constrain each part of the model?

Calibration begins with independently measured structure

Input	Measurement or proposed inference
Geometry and fluid	Puck height and area, temperature-dependent viscosity and density.
PSD and available fines	Fine fraction, representative sizes and an independently constrained releasable share.
Initial structure	CT porosity and heterogeneity. Accessible specific surface σ with resolution uncertainty.
Hydraulics	Low-flow pressure–flow slope for K_0 , empty-basket loss for R_s .
Future pore networks	Connectivity-sensitive permeability and capture/accessibility closures.

CT-derived hydraulic diameter is a structural proxy, not a direct measurement of every pore throat.

Experiments to distinguish competing kinetic models

- Fines collected in time intervals at several flows constrain release and capture.
- Interrupted CT locates retained material and redistribution.
- Pressure pulses test deposit remobilization and threshold behavior.
- Outlet size distributions test size selectivity.
- Basket-retained mass separates puck transport from cup collection.

Model comparisons One versus two retained populations; accessibility factor versus $P_{\text{mob}} = 1$; uniform capture versus axial weighting.

These are proposed experiments and comparisons. Validate on withheld pucks or grind settings.

Next comparison: redistribution versus washout

Retained redistribution Fines release and move, but deposition elsewhere in the puck dominates. CT changes while cumulative outlet mass stays small.

Net washout Mobile fines reach the puck outlet. Basket retention and cup collection quantify the outgoing mass.

Proposed study

- Compare pressure or flow levels with the same initial puck structure.
- Record M_a , M_m , M_d and M_{out} with axial deposit profiles.
- Darcy is sufficient here. Compare both permeability predictions with data.
- Use PNM or 2D/3D only where pore connectivity or radial bypass demands it.

Section 7 of 7

Conclusions and next steps

What the model shows and which evidence is still needed

Conclusions and next steps

- Fines inventories evolve through conservative transport and local exchange.
- Darcy gives pressure and flow from the evolving hydraulic resistance.
- Inertia is negligible in the tested case, while permeability closure strongly affects the predictions.
- Calibration needs independent structure, pressure–flow and fines measurements.

**The next step is to constrain the permeability and kinetic closures with data.
Physical validation remains outstanding.**

References and model documentation

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- S. Ergun, “Fluid Flow through Packed Columns,” *Chemical Engineering Progress*, 48, 89–94, 1952.
- FineFlow–Espresso technical report, Python solver, named-case configuration and automated verification suite accompanying this presentation.